



University of Asia Pacific

Department of Computer Science and Engineering

Final Project Report

CSE 438: Robotics Lab

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From Screen Time to Build Time: A Smart Device Operated Learning Robot

1. Abstract

The rapid growth of smartphone usage among children and students has significantly increased passive digital consumption through gaming, social media, and entertainment platforms. While technology provides access to information and communication, excessive passive screen time reduces opportunities for creativity, practical problem-solving, and hands-on learning. At the same time, robotics and STEM education remain inaccessible to many students due to high costs, limited infrastructure, and lack of practical educational environments, especially in rural and under-resourced institutions. This project presents a low-cost smart educational robot designed to transform passive screen time into active technological learning. The proposed system integrates Arduino-based embedded systems, Bluetooth communication, obstacle avoidance technology, motor control, voice interaction, and mobile device operation to create an engaging robotics learning platform for beginner students. The robot enables students to learn practical robotics concepts through direct interaction, experimentation, and real-world implementation. The project was designed using affordable and commonly available electronic components to ensure accessibility for both rural and urban educational institutions. Functional testing, demonstrations, and educational evaluations were conducted to validate system performance and learning effectiveness. Expert feedback and project showcase participation confirmed that the proposed system has strong potential as an affordable STEM learning platform capable of increasing student engagement, improving technological understanding, and encouraging practical robotics education.

2. Objectives

2.1 General Objective

To develop a low-cost smart educational robot that transforms passive mobile usage into active robotics and STEM learning through interactive and practical technological engagement.

2.2 Specific Objectives

1. To design and develop a Bluetooth-controlled educational robot.

2. To integrate obstacle detection and avoidance functionality using ultrasonic sensors.
3. To implement voice and sound-based interaction for user engagement.
4. To create an affordable robotics platform suitable for beginner students.
5. To encourage hands-on STEM learning in both rural and urban educational settings.
6. To increase student interest in robotics, embedded systems, and artificial intelligence.
7. To evaluate the educational effectiveness of the system through surveys, demonstrations, and exhibitions.
8. To provide a scalable robotics learning solution for workshops and classroom environments.

3. Introduction

Technology has become an inseparable part of modern life, especially among children and students. Smartphones and digital devices are now widely used from an early age. However, most interactions with these devices are passive in nature, including activities such as gaming, social media browsing, and short-form video consumption. Although digital technology offers educational opportunities, excessive passive usage limits creativity, critical thinking, and practical learning experiences.

At the same time, robotics and STEM education are becoming increasingly important in preparing students for future technological challenges. Robotics learning develops problem-solving ability, logical thinking, creativity, teamwork, and engineering skills. Despite its importance, robotics education remains inaccessible to many students due to high equipment costs, lack of infrastructure, and limited practical learning opportunities.

The situation is particularly challenging in rural and under-resourced educational institutions where access to advanced technology, laboratory facilities, and robotics training is limited. Many students learn theoretical concepts without experiencing practical implementation, making robotics and engineering appear difficult and inaccessible.

To address these challenges, this project proposes a low-cost interactive educational robot that encourages students to actively engage with technology through practical experimentation and robotics interaction. The system combines mobile device control,

obstacle detection, voice interaction, and embedded systems technology to create an engaging STEM learning environment.

The proposed robot transforms mobile phones from passive entertainment devices into active learning tools. By integrating affordable hardware components and user-friendly interaction systems, the project aims to promote inclusive robotics education for students in both rural and urban environments.

The project also emphasizes educational implementation through workshops, demonstrations, surveys, and project showcases to evaluate its learning effectiveness and social impact.

4. Proposed Methodology

The proposed educational robot was developed using a systematic engineering methodology consisting of planning, hardware development, software implementation, system integration, testing, and educational evaluation. The methodology was designed to ensure affordability, functionality, scalability, and ease of learning for beginner students.

The complete development process was divided into several phases to maintain proper workflow and implementation efficiency shows figure-1.

4.1 System Design and Planning

The first phase focused on identifying the educational problem, defining system requirements, and planning the overall architecture of the robot.

The following activities were completed during this phase:

- Identification of problems related to passive smartphone usage among students
- Analysis of affordable robotics learning solutions
- Selection of suitable hardware and software components
- Design of the system architecture and workflow

The system was intentionally designed to be:

- Low-cost
- Portable
- User-friendly

- Educational
- Easy to maintain

The robot was planned to support:

- Mobile control
- Obstacle avoidance
- Interactive learning
- Beginner robotics education

4.2 Hardware Development

Table-1 shows the hardware development phase involved designing and assembling the physical robotic system.

Component Name	Purpose
Arduino Uno	Main controller to process sensor data and control motors
Bluetooth Module (HC-05)	Enables mobile phone control & wireless commands
Motor Driver (L298N)	Controls speed and direction of DC motors
Ultrasonic Sensor (HC-SR04)	Detects obstacles and enables obstacle avoidance
Microphone Module	Detects sound/clap to trigger robot interaction
DC Gear Motors	Provide movement to the robot
Robot Wheels	Enable smooth movement and turning
Servo Motor	Rotates camera or ultrasonic sensor for scanning
Speaker	Outputs voice and sound responses
Li-ion Batteries (18650)	Power supply for the robot
Battery Holder & Protection	Holds and protects batteries safely
Charging Case	Recharges the batteries
Power Switch	Turns the robot on or off
Buzzer	Status indication and alerts
Chassis (Acrylic sheet)	Structural body of the robot
Jumper Wires	Connect all electronic components
CA-8403B Bluetooth module and audio amplifier	Audio interaction

Table-1: Hardware Components Used

Hardware Assembly Process

Step 1: Chassis Fabrication

The robot chassis was fabricated using acrylic sheets to ensure structural strength and durability.

Step 2: Motor Installation

DC gear motors and wheels were mounted onto the chassis for movement and navigation.

Step 3: Sensor Installation

The ultrasonic sensor was mounted on a servo motor to enable directional scanning during obstacle detection.

Step 4: Electronic Integration

Arduino Uno, motor driver, Bluetooth module, speaker, buzzer, and microphone module were integrated into the system.

Step 5: Power System Setup

Li-ion batteries and battery holders were connected to provide portable power supply. We give total 11.1 volt (3 Li-ion battery of 3.7v).

4.3 Software Development

The software system was developed using the Arduino IDE platform. Embedded programming techniques were used to control sensors, motors, communication modules, and interaction systems.

The software implementation focused on:

- Robot navigation
- Motor control
- Obstacle avoidance

- Bluetooth communication
- Sound-based interaction
- Real-time response processing
- Functionalities Implemented

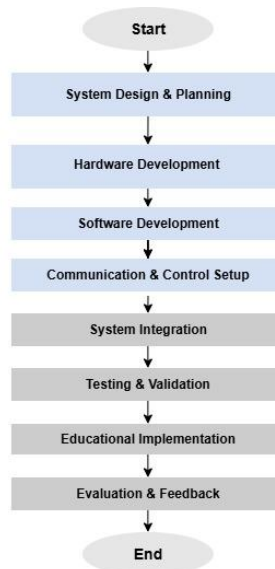


Figure-1: Overall Workflow of the Proposed Educational Robot System

Motor Control System

The robot supports:

- Forward movement
- Backward movement
- Left turning
- Right turning
- Stop functionality

Obstacle Avoidance System

The ultrasonic sensor continuously measures the distance between the robot and nearby obstacles. If an object is detected within a predefined safe distance:

1. The robot stops.
2. The servo motor rotates the sensor.
3. Left and right distances are measured.
4. The robot selects the safest direction.

Bluetooth Communication

The HC-05 Bluetooth module enables wireless communication between the robot and a smartphone.

Sound-Based Interaction

The microphone module detects sound or clap signals to trigger interactive responses.

Alert System

The buzzer and LEDs provide warning and system status indications.

4.4 Pseudocode

The following pseudocode describes the operational logic of the robot system.

Start

Initialize Arduino pins

Initialize Bluetooth module

Initialize ultrasonic sensor

Initialize motor driver

Initialize servo motor

Initialize buzzer

Set operating modes:

Manual control mode

Obstacle avoidance mode

Interactive mode

Loop continuously

If Bluetooth command is received:

If command = Forward:

Move robot forward

Else if command = Backward:

Move robot backward

Else if command = Left:

Turn robot left

Else if command = Right:

Turn robot right

Else if command = Stop:

Stop robot

If obstacle avoidance mode is active:

Measure front distance

If front distance > safe distance:

Move robot forward

Else:

Stop robot

Turn buzzer ON

Rotate servo left

Measure left distance

Rotate servo right

Measure right distance

If left distance > right distance:

Turn robot left

Else:

Turn robot right

If sound or voice signal is detected:

Trigger interactive response

Repeat continuously

End

4.5 Communication and Control System

A Bluetooth-based wireless communication system was established between the robot and a smartphone.

Mobile Control Features

The user can:

- Navigate the robot manually
- Send movement commands
- Control robot behavior wirelessly
- Interact with the robot through a mobile device

The communication system was implemented using:

- HC-05 Bluetooth module
- Smartphone Bluetooth application
- Serial communication protocol

This system transforms smartphones from passive entertainment devices into active robotics learning tools.

4.6 System Integration

In this phase, all hardware and software modules were combined into a unified robotic system.

Integration Activities

- Connecting all sensors and actuators
- Synchronizing hardware and software components
- Establishing communication between modules
- Testing motor and sensor coordination
- Ensuring stable power distribution

The following subsystems were integrated:

- Motor control system
- Obstacle avoidance system
- Bluetooth communication system
- Audio interaction system
- Power management system

Real-time synchronization was optimized to improve system stability and performance.

4.7 Testing and Validation

Extensive testing and validation were conducted to evaluate the functionality, reliability, and educational usability of the robot.

Functional Testing

- The robot's movement accuracy was tested in:
 - Forward movement
 - Backward movement
 - Left turning
 - Right turning
- The ultrasonic sensor was tested to ensure accurate obstacle detection and automatic avoidance behavior.
- Wireless command transmission between smartphone and robot was verified.
- Servo rotation and directional scanning performance were evaluated.
- The microphone module was tested for clap and sound-triggered interaction.
- Battery performance and system stability were monitored during operation.

Validation Criteria

The robot was evaluated based on:

- Navigation accuracy
- Sensor responsiveness
- Wireless communication reliability
- System stability
- Ease of use
- Educational effectiveness

The system successfully passed all major functionality tests and demonstrated stable performance during implementation and exhibition activities.

4.8 Educational Implementation and Evaluation

The final phase focused on evaluation of the proposed system.

Evaluation Methods

Feedback was collected through:

- Expert surveys
- Offline discussions
- Demonstration observations
- Project exhibition feedback

The evaluation confirmed that the proposed educational robot:

- Improves student engagement
- Encourages practical STEM learning
- Increases interest in robotics and technology
- Supports beginner-level robotics education

The project also received positive recognition during robotics exhibitions and project showcase competitions.

5. Social Impact

The proposed educational robot creates both educational and social benefits.

5.1 Educational Impact

- Encourages practical STEM learning.
- Improves understanding of robotics concepts.
- Enhances problem-solving ability.
- Promotes creativity and innovation.
- Supports beginner robotics education.

5.2 Social Impact

- Reduces passive screen time among students.
- Encourages productive technology usage.
- Provides affordable robotics learning opportunities.
- Promotes technological awareness in rural communities.
- Increases student engagement in engineering and AI.

5.3 Economic Impact

- Low-cost implementation (~\$5,000).
- Affordable for schools and workshops.
- Scalable for mass educational deployment.

6. Budget

6.1 Budget Table

Sl. No.	Item Name	Qty	Cost/unit (BDT)	Subtotal (BDT)
1	Arduino Uno (Compatible)	1	950	950
2	Ultrasonic Sensor (HC-SR04)	1	110	110
3	Bluetooth Module (HC-05)	1	349	349
4	CA-8403B Bluetooth module and audio amplifier	1	300	300
5	L298N Motor Driver	1	185	185
6	DC Gear Motor	4	150	600
7	Robot Wheels	4	80	320
8	Servo Motor (SG90)	1	350	350
9	Speaker	1	150	150
11	Li-ion Battery (18650)	3	120	360
12	Battery Holder	1	180	180
13	Battery Charging Case	1	220	220
15	Acrylic Sheet	1	700	700
16	Jumper Wires	1 set	80	80
17	Power Switch	1	10	10
18	Buck converter LN2596	1	100	100
19	Buzzer (1 pc)	1	30	30

Table-2 : Budget Table

Table-2 shows total budget: Five thousand taka only

Total budget ~5,000 taka

7. Circuit Diagram

Circuit Description

Figure-2 shows the Arduino Uno acts as the central controller of the system. The L298N motor driver controls the movement and speed of the DC motors. The HC-SR04 ultrasonic sensor detects obstacles, while the servo motor rotates the sensor to scan surrounding directions.

The HC-05 Bluetooth module enables wireless communication between the robot and a mobile phone. The buzzer provide alerts and status indications.

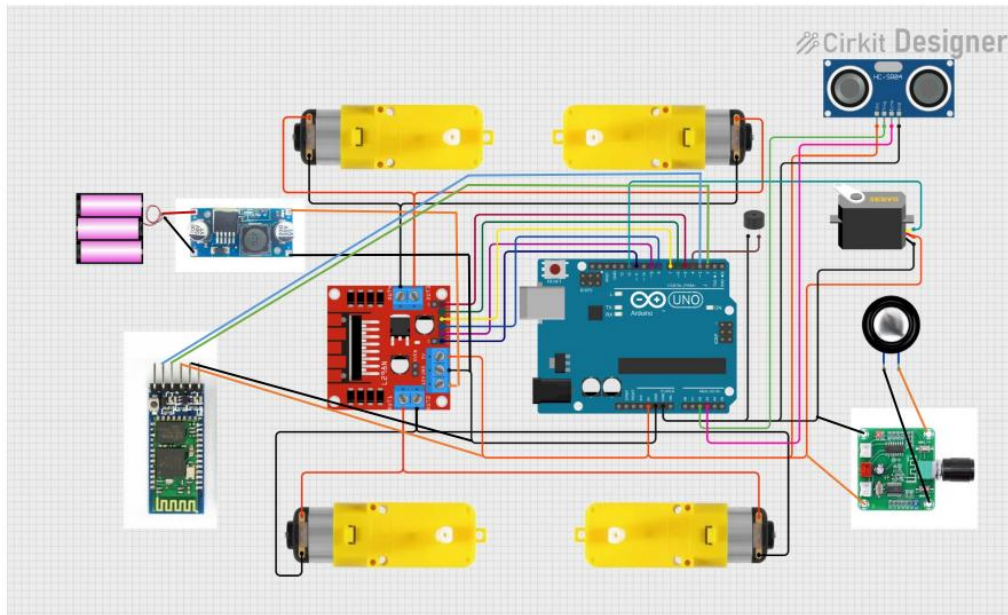


Figure-2 Circuit diagram

Main Connections

Component	Arduino Pin
HC-05 TX	3
HC-05 RX	2
Ultrasonic Trigger	A3
Ultrasonic Echo	A2
Servo Motor	12
Motor Driver IN1-IN4	D6-D9
Motor Enable Pins	D5 & D11
Buzzer	4

8. Test / Implementation

8.1 Chassis Fabrication

The robot chassis was first designed digitally using AutoCAD software to create an accurate structural layout for component placement and overall body dimensions. The design included motor positioning, wheel alignment, sensor placement, and mounting

space for electronic modules. After completing the design, the acrylic sheets were cut according to the AutoCAD dimensions to fabricate the robot body. The chassis was designed using acrylic sheets to create a lightweight yet durable structure capable of supporting all hardware components. Wheels and DC motors were mounted securely to ensure smooth movement and balanced navigation during operation.

During fabrication, alignment issues occurred while positioning the motors and wheels, which initially affected robot balance and movement accuracy. Multiple adjustments were made to improve wheel alignment, chassis stability, and proper fitting of the electronic components.

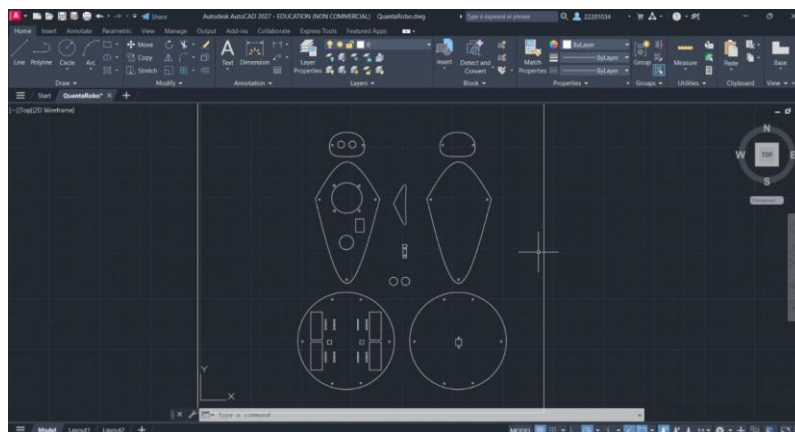


Figure-3: Autocad design

8.2 Hardware Assembly

After completing the chassis fabrication, all electronic components including the Arduino Uno, ultrasonic sensor, motor driver, Bluetooth module, servo motor, microphone module, speaker, and buzzer were assembled onto the robot body. Components were positioned carefully to reduce wiring complexity and improve accessibility during debugging.

One major challenge during this phase was limited space management inside the chassis, especially while arranging wires and modules. Additional adjustments were required to avoid loose connections and hardware interference during movement.



Figure-4: Main hardware of the system

8.3 Circuit Wiring

All hardware modules were connected according to the designed circuit diagram. Power distribution, signal connections, and communication wiring were completed carefully to ensure stable operation.

During testing, several issues related to loose jumper wire connections and incorrect pin configurations caused unstable system behavior. Repeated wiring verification and debugging were performed to resolve these problems and improve system reliability.

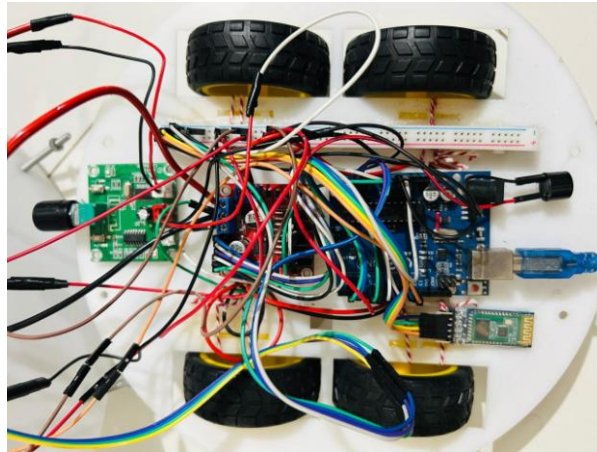


Figure-5: Full circuit setup

8.4 Programming

The software system was developed using Arduino IDE. Programming focused on motor control, obstacle avoidance, Bluetooth communication, servo scanning, and interactive functionalities. The code was uploaded to the Arduino Uno and tested continuously during implementation. One of the main challenges faced during programming was integrating multiple functionalities such as Bluetooth control, obstacle avoidance, and sound interaction into a single system. Initially, combining these modules caused delays and unstable behavior. Several iterations of debugging and code optimization were performed to achieve synchronized real-time operation.

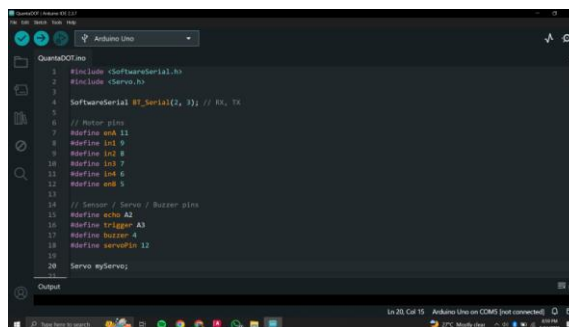


Figure-6: Coding in Arduino IDE

8.5 Obstacle Detection Testing

The ultrasonic sensor was tested to evaluate obstacle detection accuracy and autonomous navigation capability. The sensor successfully detected nearby objects, stopped the robot, and triggered servo-based directional scanning to identify safer movement paths.

Initially, inaccurate distance measurements and delayed servo responses affected obstacle avoidance performance. Sensor positioning and timing delays in the code were later optimized to improve detection accuracy and movement decisions.



Figure-7: Obstacle detect and avoiding part

8.6 Bluetooth Control Testing

Wireless robot navigation was tested using a smartphone Bluetooth application connected through the HC-05 Bluetooth module. The robot successfully responded to forward, backward, left, right, and stop commands in real time. At the beginning of testing, unstable Bluetooth pairing and communication delays occasionally interrupted control signals. After adjusting baud rate settings and improving power stability, the wireless communication system operated reliably.

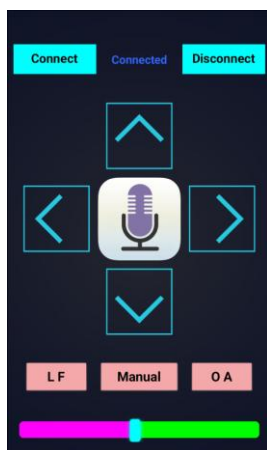


Figure-8: Bluetooth & voice control application

8.7 Final Demonstration

The completed robot was demonstrated project showcase events, and robotics exhibitions. The system successfully performed mobile-controlled navigation, obstacle avoidance, and interactive operation in front of participants and evaluators. The project received positive responses for its affordability, innovation, and educational impact, and later achieved recognition in project showcase competitions.



Figure-9: Participation in project showcase event

[Our project short demo YouTube link](#)

9. Gantt Chart

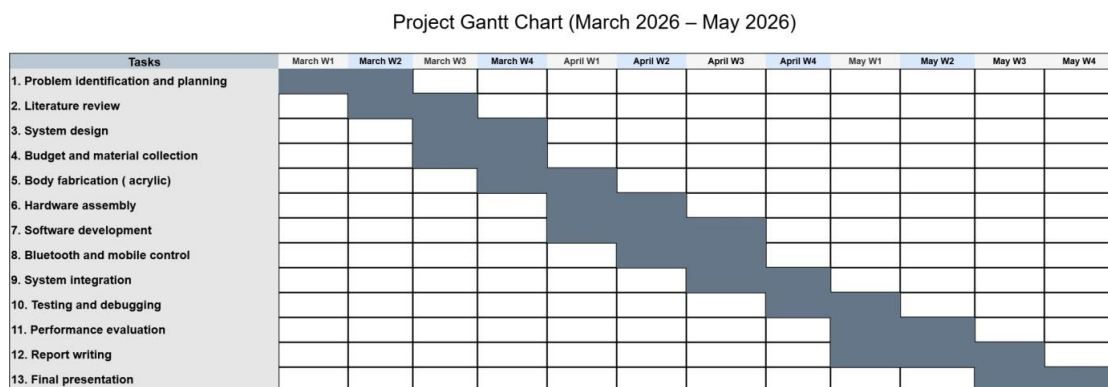


Figure-10 : Project timeline

Figure-10 shows project timeline :

Task	Timeline	Short Description
1. Problem Identification and Planning	March W1 – March W2	The project problem was identified by analyzing excessive passive screen time among students and the lack of affordable robotics education. Initial planning, goal setting, and project roadmap preparation were completed during this phase.
2. Literature Review	March W2 – March W3	Existing educational robotics systems, embedded technologies, mobile-controlled robots, and STEM learning approaches were researched to improve the project design and innovation.
3. System Design	March W3 – March W4	The overall architecture of the robot was designed, including hardware layout, software workflow, component selection, circuit planning, and interaction system design.
4. Budget and Material Collection	March W3 – March W4	A low-cost budget plan was prepared and required materials such as Arduino, sensors, motors, batteries, and acrylic sheets were collected for implementation.
5. Body Fabrication (Acrylic)	March W4 – April W1	The robot chassis and body structure were fabricated using acrylic sheets through cutting, shaping, and assembly to ensure durability and proper component placement.
6. Hardware Assembly	April W1 – April W2	Electronic components including motors, motor drivers, ultrasonic sensors, Bluetooth module, buzzer, and power supply systems were assembled and connected.
7. Software Development	April W1 – April W3	Embedded software was developed using Arduino IDE to implement motor control, obstacle avoidance, Bluetooth communication, sound interaction, and autonomous functionalities.
8. Bluetooth and Mobile Control	April W2 – April W3	Wireless communication between the robot and smartphone was established using the HC-05 Bluetooth module to enable remote navigation and interactive mobile control.
9. System Integration	April W3 – April W4	All hardware and software subsystems were integrated into a unified robotic platform to ensure synchronization, stable communication, and real-time operation.
10. Testing and Debugging	April W4 – May W1	Functional testing was conducted to evaluate movement accuracy, obstacle detection, Bluetooth

		responsiveness, and system stability. Errors were identified and corrected through debugging.
11. Performance Evaluation	May W1 – May W2	The overall system performance, educational effectiveness, usability, and interaction quality were evaluated through demonstrations and practical operation.
12. Report Writing	May W1 – May W3	Technical documentation including methodology, implementation, circuit design, survey findings, testing results, and analysis was compiled into the final professional report.
13. Final Presentation	May W3 – May W4	The completed project was demonstrated in robotics exhibitions and project showcases, where the system functionality, educational value, and innovation were presented to judges and participants.

Table-3 : Project timeline description

10. Future Work

Future improvements of the proposed educational robot may include the integration of an AI-based voice assistant to enable more natural human–robot interaction and intelligent response systems. A dedicated mobile application can be developed to provide a more user-friendly and interactive control interface for students. The addition of a camera and computer vision system would allow the robot to perform object detection, environmental analysis, and smart navigation tasks. Gesture control functionality could further enhance interactive learning by enabling touchless robot operation. Future versions may also incorporate IoT and cloud connectivity to support remote monitoring, online data exchange, and smart educational features. A face recognition system could be implemented for personalized interaction and user identification. To improve sustainability and portability, solar-powered charging mechanisms may be integrated into the system. An OLED-based emotion display could make the robot more expressive and engaging for younger learners. Online robotics learning modules and tutorials may also be connected with the platform to support self-learning and remote STEM education. Additionally, advanced autonomous navigation algorithms can be implemented to improve intelligent movement, path planning, and real-time obstacle avoidance capabilities.

11. Conclusion

This project successfully demonstrates the development of a low-cost smart educational robot capable of promoting practical robotics learning and STEM education. The system integrates mobile control, obstacle avoidance, embedded systems, and interactive technologies to create an engaging educational platform for students.

The robot transforms passive smartphone usage into productive technological interaction and encourages hands-on experimentation. Due to its affordability and modular design, the system can be implemented in schools, workshops, and beginner robotics learning environments.

Survey evaluation, demonstrations, and exhibition participation confirmed the educational value and practical applicability of the project. The achievement of champion status in project showcases further validated the innovation and effectiveness of the proposed system.

With future improvements and scalability, the project has strong potential to contribute to accessible robotics education and technological development in both rural and urban educational institutions.

12. References

1. Arduino Official Documentation
2. HC-SR04 Ultrasonic Sensor Datasheet
3. HC-05 Bluetooth Module Manual
4. L298N Motor Driver Documentation
5. Embedded Systems Design References
6. Robotics and STEM Education Research Articles
7. IEEE Educational Robotics Publications
8. Mobile-Controlled Robot Research Papers
9. Arduino IDE Programming Guide
10. Educational Technology and STEM Learning Journals

13. Appendix

Appendix A: Survey Report Summary



Figure-11: Offline Expert Evaluation and Project Discussion Session

Survey and Expert Evaluation

To evaluate the educational value and practical feasibility of the proposed robot, an expert survey and offline discussion shows in figure-11 session were conducted with Dr. Nazmun Nahid, Assistant Professor of the Department of Computer Science and Engineering at the University of Asia Pacific. The evaluation focused on the project concept, system architecture, educational effectiveness, and overall innovation of the proposed robotics platform. A structured questionnaire containing close-ended, mixed-type, and open-ended questions was used to collect feedback regarding the robot's functionality, usability, and learning impact. During the offline discussion session, the project workflow, system implementation, and educational objectives were demonstrated in detail. The expert evaluation indicated that the project concept is meaningful, innovative, and highly suitable for beginner robotics education. The respondent particularly appreciated the AI-inspired interaction system, mobile-controlled learning approach, and practical STEM learning potential. Recommendations were also provided for improving robot interactivity, adding expressive output systems, and developing future educational workshop modules.

The survey findings confirmed that the proposed educational robot has strong potential to transform passive screen time into productive technological learning and can effectively support hands-on robotics education in both rural and urban learning environments.

Appendix B: Robo Expo & Project Showcase Achievement

The project was demonstrated at a robotics exhibition and project showcase competition. During the event, the robot attracted significant attention due to its educational purpose, interactive operation, and low-cost implementation.

The team successfully achieved the **Champion Award** in the project showcase competition. The judges appreciated:

- Educational innovation
- Practical implementation
- Interactive learning concept
- Affordable design approach

The achievement validated the technical quality and educational significance of the project.



Figure-12: Moments of RoboExpo 3.0 participation and achieved the **Champion Award**

Appendix C: CEP Mapping

How Ks are addressed through the Project

Ks	Attributes	How Ks are Addressed through the Project	CLOs	PLOs
K3	A systematic, theory-based formulation of engineering fundamentals	The project applies fundamental concepts of electronics, programming, and control systems using Arduino, motor drivers, and sensors to develop an educational robot.	CLO1, CLO2	PLO1
K4	Engineering specialist knowledge with theoretical frameworks	The system integrates embedded programming and wireless communication (Bluetooth) based on standard engineering frameworks.	CLO2, CLO3	PLO2
K5	Knowledge that supports engineering design in practice	The project involves practical design and implementation of a low-cost robotic system, including circuit design, component integration, and system architecture.	CLO3	PLO3
K6	Knowledge of engineering practice (technology)	Real-world technologies such as Arduino Uno, motor drivers, ultrasonic sensors, microphone modules, and mobile control are used to develop a functional prototype.	CLO3, CLO4	PLO4

How As are addressed through the Project

As	Attributes	How As are Addressed through the Project
A1	Range of resources	The project utilizes hardware components (Arduino, motors, sensors), software tools (Arduino IDE), and mobile-based control interfaces.
A2	Level of interaction	Users interact with the robot through mobile control (Bluetooth) and sensor-based responses, enabling active learning.

A5	Familiarity	The system uses simple and commonly available technologies, making it suitable for primary-level students in both rural and urban settings.
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How Ps are addressed through the Project

Ps	Attribute	How Ps are Addressed through the Project	CLOs	PLOs
P1	Depth of knowledge required	The project requires understanding of embedded systems, sensor integration, programming, and hardware-software interaction.	CLO2, CLO3	PLO2
P2	Range of conflicting requirements	The design balances cost, usability, and performance to make the robot affordable and suitable for rural and urban primary-level students.	CLO3	PLO3
P7	Interdependence	The project combines multiple subsystems such as hardware assembly, software programming, communication modules, and sensors, demonstrating interdependent functionality.	CLO3, CLO4	PLO4

